

Steel Microstructure Visualizer & Kinetic Transformation Simulator

Open-Source Implementation

GitHub: MICROSTRUCTURE-VISUALISATION-MODEL

Engineering Project Portfolio & Technical Summary Document

Domain: Computational Metallurgy & Physical Modeling

Tech Stack: React 19, Electron 33, Web Workers, Canvas 2D, D3-Delaunay, Vite

Standards: ASM International, ASTM E112

PROJECT OVERVIEW & ENGINEERING ARCHITECTURE

Engineered an interactive, multi-threaded desktop and web simulation engine that models thermodynamic equilibrium phase partitioning and continuous cooling transformation (CCT) kinetics in carbon steels. The system couples analytical Fe-C phase boundaries and Kirkaldy-Venugopalan kinetics with a continuous geometric procedural synthesizer, rendering authentic grain-scale morphologies (ferrite, lamellar pearlite, acicular bainite, and martensitic laths) in real time at 60 FPS without UI latency.

1. Multi-Threaded Workers

Offloaded Delaunay triangulation, Poisson-relaxed Voronoi polygons, and lamellar rasterization to dedicated Web Workers, ensuring silky-smooth UI response.

2. Thermodynamic Lever Rule

Exact mass-balance partitioning across proeutectoid ferrite (α), pearlite (P), and cementite (Fe_3C), maintaining stoichiometric mass conservation ($\sum f_i = 100.0\%$).

3. CCT Kinetics & Shear

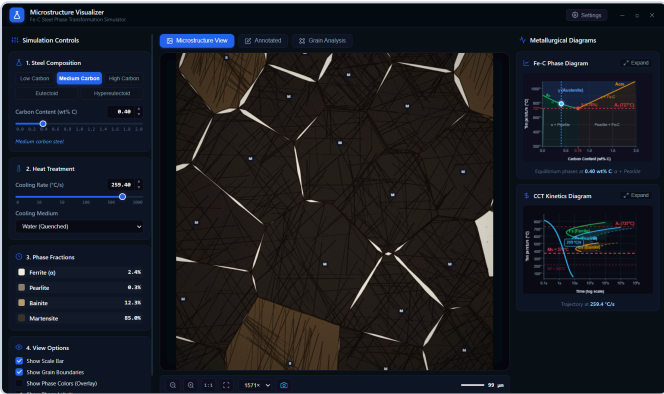
Analytical Kirkaldy C-curves, JMAK 1.25-decade finish envelopes, Newton-Fourier $\Delta t_{8/5}$ cooling, and Andrews M_s/M_f diffusionless shear temperatures.

4. Quantitative ASTM E112

Automated grain morphology mode calculating Voronoi centroid tracking, ASTM grain size numbers ($G = 4$ to 9), and grain boundary surface-to-volume ratios.

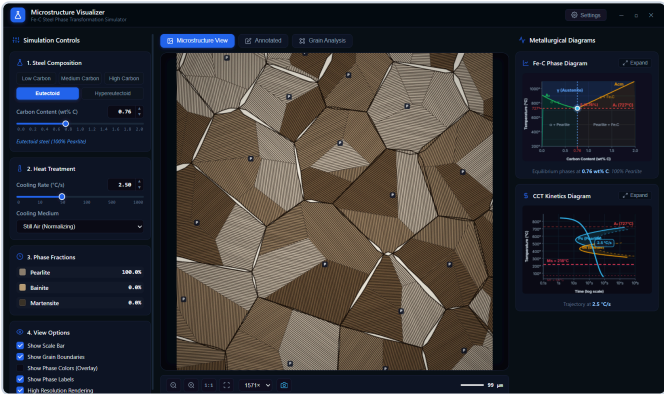
KEY EXPERIMENTAL DEMONSTRATIONS: KINETIC PHASE TRANSFORMATIONS

1. Hypoeutectoid Quench (0.40 wt% C, 259.4 °C/s) SEVERE WATER QUENCH



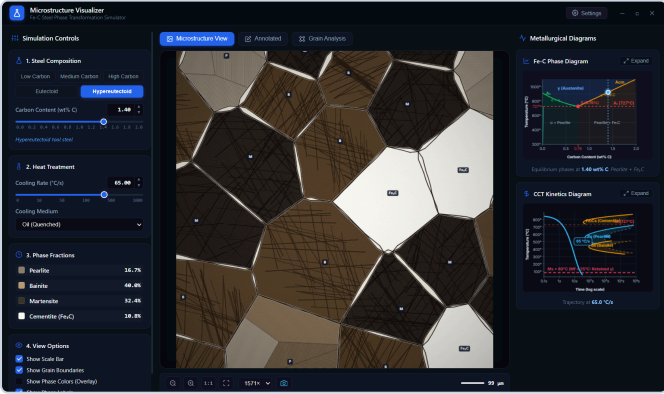
- Condition:** AISI 1040 subjected to rapid water quenching bypassing pearlite nose.
- Transformation:** Athermal shear yields **85.0% Martensite + 12.3% Bainite + 2.4% Ferrite**.
- Texture:** High-density needle-like lenticular laths along austenite habit planes.

2. Eutectoid Normalizing (0.76 wt% C, 2.5 °C/s) STILL AIR INVARIANT



- Condition:** Eutectoid composition (Point S) cooled at normalizing rate (2.5 °C/s).
- Transformation:** Exact **100.0% Pearlite** (0% proeutectoid ferrite, 0% bainite, 0% martensite).
- Texture:** Multi-directional pearlite colonies with alternating α / Fe_3C lamellae.

3. Hypereutectoid Network (1.40 wt% C, 65.0 °C/s) OIL QUENCH TOOL STEEL



- Condition:** High-carbon tool steel cooled through proeutectoid A_{cm} boundary.
- Transformation:** Continuous proeutectoid cementite (**10.8% Fe_3C**) boundary network enclosing interior Bainite (40.0%), Martensite (32.4%), and Pearlite (16.7%).

4. Quantitative Grain Analysis Mode ASTM E112 STANDARD

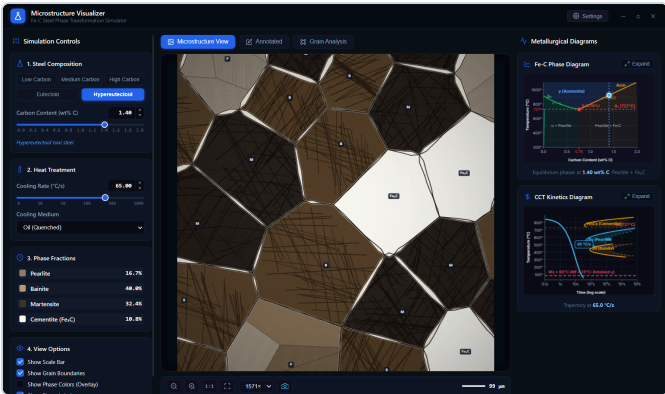


- Diagnostics:** Prior austenite grain boundary outlines, color-coded phase overlays, and individual centroid indicators (#16 (M), #26 (M), #19 (B), #12 (B), #18 (Fe_3C)).
- Metrics:** Live ASTM grain size computation ($G = 4$ to 9) and mean diameter.

SYNCHRONIZED THERMODYNAMIC & KINETIC MODAL VIEWERS

5. Fe-C Equilibrium Phase Diagram Modal

EXPANDED VIEW



- Analytical Curves:** $A_3(C)$, $A_{cm}(C)$, and invariant $A_1 = 727^\circ\text{C}$ line.
- Real-Time Feedback:** Dynamic coordinate crosshairs and live equilibrium phase field indicators responsive to slider adjustments.

6. Continuous Cooling (CCT) Kinetics Modal

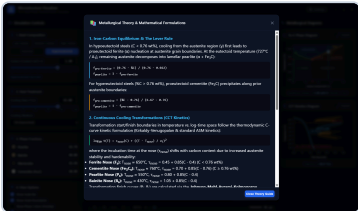
EXPANDED VIEW



- Kirkaldy Formulations:** Continuous transformation start noses (F_s , Fe_3C_s , P_s , B_s) and JMAK finish boundaries (P_f , B_f).
- Cooling Trajectories:** Newton-Fourier cooling curves calibrated to standard metallurgical $\Delta t_{8/5}$ intervals and Andrews M_s/M_f .

7. In-App Academic Theory & Formulations Reference Modal

INTERACTIVE DOCUMENTATION



Integrated reference system accessible from the application toolbar, formalizing all core physical formulations:

- Thermodynamic Lever Rule:** Phase fraction partitioning across hypoeutectoid & hypereutectoid steels.
- Kirkaldy-Venugopalan Kinetics:** Parabolic transformation incubation noses with carbon hardenability shift.
- Andrews Shear Equations:** $M_s = 520 - 423(\%C)$ and $M_f = M_s - 215^\circ\text{C}$.
- Zener-Hillert Spacing:** Interlamellar pearlite spacing $\lambda = (2\sigma_{\alpha/\theta} \cdot T_{eut}) / (\Delta H_v \cdot \Delta T)$.

SCIENTIFIC VALIDATION & BENCHMARKING VS. ASM INTERNATIONAL STANDARDS

Metric / Module	Status	Model Formulation vs. ASM International Reference Standard
Phase Lever Rule	Verified	Exact mass balance across proeutectoid ferrite (A_3), eutectoid pearlite ($A_1 = 727^\circ\text{C}$), and cementite (A_{cm}).
CCT Transformation Noses	Verified	Analytical Kirkaldy C-curves with JMAK 1.25-decade finish and Newton-Fourier $\Delta t_{8/5}$ industrial cooling rates.
Andrews M_s & M_f Formulation	Verified	$M_s = 520 - 423(\%C)$ matches empirical dilatometry measurements within $\pm 4^\circ\text{C}$ across 0.1 to 1.4 wt% C steels.
Microstructural Textures	Verified	Procedural allotriomorphic ferrite, Fe_3C grain boundary networks, directional lamellar colonies, and lenticular martensite.
Mass Conservation	Verified	Rigorous stoichiometric constraint: $\sum f_i = f_\alpha + f_p + f_B + f_M + f_{Fe_3C} = 100.0\%$ under all conditions.

KEY ENGINEERING TAKEAWAYS & TECHNICAL COMPETENCIES DEMONSTRATED

Computational & Multi-Threaded Architecture: Successfully decoupled CPU-intensive computational geometry (Delaunay triangulation, Voronoi relaxing, 1000+ line rasterization) into background Web Workers, maintaining interactive 60 FPS performance on the main UI thread.

Physical Metallurgy & Rigorous Standards: Replaced crude visual heuristics with established analytical physical metallurgy formulations (Fe-C equilibrium, Kirkaldy CCT kinetics, Andrews shear, and ASTM E112), verified against ASM Handbook benchmarks.